East Swindon Development Group

Eastern Villages Swindon (South Marston)

FRAMEWORK LANDSCAPE, ECOLOGY AND ARBORICULTURAL MANAGEMENT PLAN (FLEAMP)

April 2023

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1.0 INTRODUCTION

1.1 This Framework, Landscape, Ecology and Arboricultural Management Plan (FLEAMP) has been produced by FPCR Environment and Design Ltd. on behalf of the East Swindon Development Group.

1.2 It provides an overarching framework management plan for the development at South Marston, in the context of, and with reference to the wider Eastern Villages development site (planning reference S/OUT/13/1555, condition 10). The planning consent is for outline permission

'for a sustainable urban extension to the east of Swindon and north of the A420 of up to 2,380 dwellings together with a mixed use local centres/areas (including A1 retail up to 1,500 sq metres, services (A2), restaurants, pubs and takeaways (A3, A4, A5), business uses (B1) up to 1,000 sq metres, residential); community uses (D1); sheltered and/or care accommodation (C2/C3); two primary schools; green infrastructure including formal (including playing fields) and informal open space, landscape, biodiversity and amenity space; play space (including NEAPS/LEAPS/MUGAS); changing and sports facilities (including D2) and ancillary retail uses; sustainable drainage systems; highway, cycle and pedestrian routes; car parking; infrastructure (including utilities); engineering works including ground remodelling; demolition, site reclamation and removal of structures; the formation of new accesses from the A420, Old Vicarage Lane and Thornhill Road. | Land North Of A420 Eastern Villages Swindon'

1.3 The FLEAMP is required to protect ecological receptors on Site and discharge Planning Condition 10 of the outline consent which states

10. Framework, Landscape, Ecology and Arboricultural Management Plan (FLEAMP) Prior to the submission of the first reserved matters at South Marston or Rowborough, a Framework Landscape, Ecology and Arboricultural Management Plan (FLEAMP) for that village in broad accordance with the Biodiversity Enhancement Options Plan (DN 1796-L03 D) (submitted to LPA on 7th September 2017) shall have been submitted to and approved in writing by the Local Planning Authority. The FLEAMPs shall provide an overarching management plan for the development at South Marston and Rowborough. The FLEAMPs shall:

a) Identify the feature specific objectives to achieve the overarching aim;

b) Provide measures to enhance the biodiversity of South Marston Brook through appropriate management of bank side vegetation, the re-grading of bank profiles and reconnecting the South Marston Brook with its floodplain to create greater morphological diversity and a range of floodplain features such as scrapes and wetlands;

c) Provide the location, principles of design, purpose, function, buffer zone and construction of the wet features (ponds and SuDS), including connectivity of water features and designs which minimise the disturbance of wetland and riverine habitats, particularly watervoles;

d) Provide a strategy for net biodiversity gain and enhancement and new priority habitat creation including advanced mitigation and protective measures during construction, long term protection objectives, habitat connectivity, a maintenance schedule and proposals for allowing and restricting public access;

e) Identify the additional green infrastructure that will be created. The development shall be laid out and made available for use in accordance with the approved FLEAMPs. The submitted FLEAMPs shall be in accord with the approved Open Space Parameter Plan.

Reason: To ensure biodiversity, green infrastructure and assets are protected, integrated with the development and enhanced in accordance with Policies SD3, EN1 and NC3 of the Swindon Borough Local Plan 2026. The FLEAMPs shall also ensure the development provides access to nature to promote human well-being and afford educational opportunities.

1.4 This FLEAMP provides a framework for South Marston (shaded pink in Figure 1 below) only. A separate FLEAMP will detail the management for the remainder of the development site (Rowborough, shaded blue).

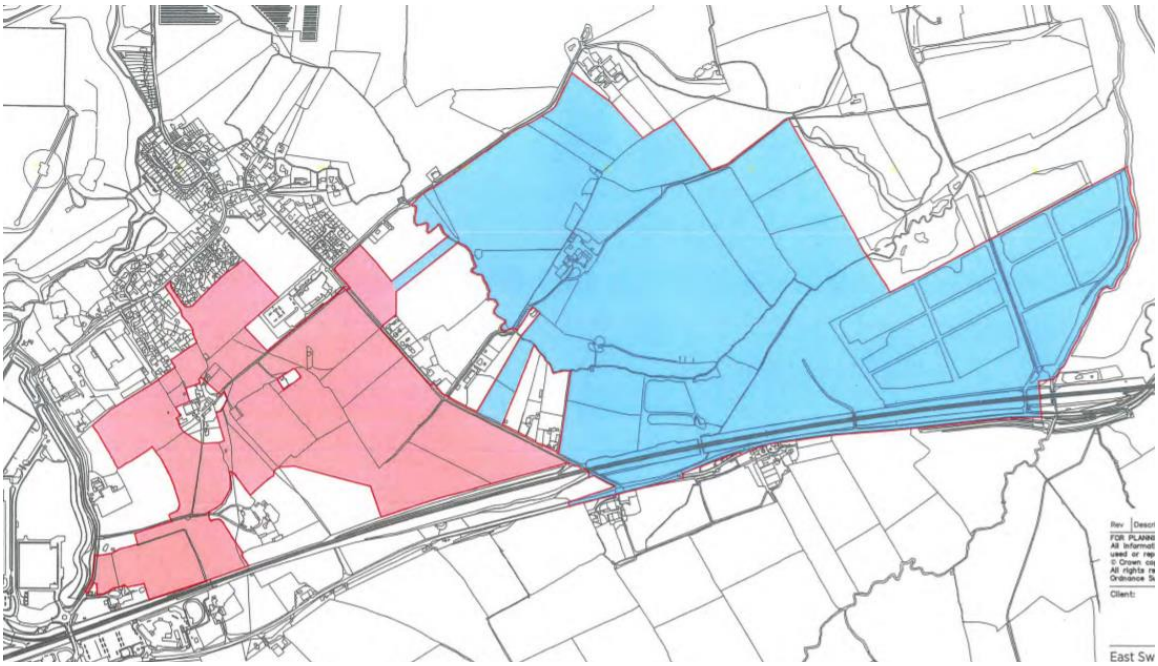


Figure 1: South Marston and Rowborough FLEAMP areas.

1.5 The FLEAMP provides a strategy for how biodiversity, green infrastructure and natural assets are to be both protected and integrated within the development, managed and enhanced in accordance with Policies SD3, EN1 and NC3 of the Swindon Borough Local Plan 2026. In addition, the FLEAMP also details how the development will provide access to nature to promote human well-being and afford educational opportunities.

Site Context

Landscape

1.6 In broad terms, the site forms a parcel of agricultural land lying between the small village of South Marston to the north and the A420 and the Great Western railway line to the south.

1.7 Although predominantly an agricultural landscape, the site lies close to the urban edge of Swindon and there are a number of urbanising elements that lie near to the site, or in close proximity which have a strong urban influence on the landscape. This includes, amongst others, the A419 (Swindon by-pass), A419-A420 interchange and large buildings at St Margaret's Retail Park to the west; the extensive South Marston Industrial Estate and Keypoint Distribution Centre to the north and west which includes a number of large employment buildings and distribution warehouses, and the railway and A420 which run parallel with one another to the south. The site boundary is defined by

a combination of the curtilage of farms and properties, hedgelines, public rights of way, roads and by the edge of South Marston. South Marston village form the site's northern boundary. The southern edge of the village is defined by residential properties, South Marston primary school, new residential development of the site of the former Mercure South Marston Hotel and Leisure Club, as well as an industrial/employment area off Thornhill Road.

1.8 The site is not subject to any statutory landscape quality designations, nor is the local landscape. Within the wider study area is the North Wessex Downs AONB to the south, which lies around 4km from the site's closest point, and the Cotswolds AONB around which lies to the north of Swindon, approximately 13km from the site at its closest point.

1.9 Conservation Areas are located in Lower and Upper Wanborough and Hinton Parva on the North Wessex Downs escarpment, Stanton Fitzwarren and Sevenhampton to the north, and within Stratton St Margaret (in Swindon) to the west. In close proximity to the site are the farmhouses of Priory Farm, Manor Farm, and Nightingale Farm which are Grade II Listed, as is the property of The Red House.

Land Use

1.10 The site is currently in agricultural use and comprises arable fields with some improved pasture for cattle grazing. The field pattern is typical of the wider landscape with fields being generally irregular in shape and predominantly of small scale, although there are some bigger fields to the east near Rowborough Farm

1.11 Hedgerows and ditches define the field pattern. Many of the hedges are tall and thick containing a number of mature trees, although some are variable in quality, largely due to current management practices. As well as some wet ditches associated with hedgelines, there are some small field ponds within the site. Broad leaved plantation woodland occurs in the form of recently planted, willow dominated woodland. Some coniferous planting also occurs.

1.12 There are a number of Public Rights of Way (PROW) that cross the site, mostly in a northsouth direction between South Marston and the A120. A Public Bridleway heads into Nightingale Wood via Rowborough Farm.

Topography

1.13 The main part of the site is relatively flat in character. The site varies between 100-90m Above Ordnance Datum with the higher part of the site at Nightingale Wood to the east. The site gently falls in a south easterly towards the Marston Brook and the River Cole.

1.14 The landform begins to gently rise to the north of South Marston at Stanton Fitzwarren (124m AOD) and Reddown Hill (135m AOD), before reaching higher land of the Cotswolds AONB to the north. To the south of the site there are some small occasional rises at Mouth Pleasant Farm (103m AOD) and at Bourton (105m AOD), but in the main the landscape is characterised by the wide and generally flat Cole Valley (90-95m AOD). Steeply rising slopes occur on the edge of the North Wessex Downs Scarp with Upper Wanborough at 140m AOD. Charlbury Hill, near Hinton Parva reaches 253m AOD. 10

Biodiversity

1.15 A range of habitats are present within the site and are typically heavily influenced by the current agricultural management of the site. The site is dominated by arable fields and improved pasture divided by hedgerows. There are occasional species-poor, semi-improved grasslands present in association with field boundaries and areas of less intensive management. Mature trees are frequently associated with hedgerows. Significant parts of the site are planted with willow coppice, which, it is understood, was planted as a source of biofuel, although remains un-harvested. Semi-improved grassland is generally highly localised within the site and persists where intensive agricultural management is reduced. All are species–poor in nature and where ungrazed, such as adjacent to Nightingale Lane, are dominated by a range of common coarse grasses. Small patches of marshy grassland occur to the south of South Marston, where damp depressions/ ditches occur. A suite of grassland species typical of water-logged conditions were prevalent. Tall ruderal vegetation is located frequently along hedgerows bases as is typical where arable and improved grassland predominate.

1.16 The site has been subject to ongoing ecological surveys dating back to 2002. The most recent protected species surveys (badgers, bats, birds, dormice and reptiles) were conducted by FPCR in 2022 covering the South Marston area. The results of these surveys are detailed in separate reports, and have been used to inform the site assessment and proposals.

2.0 POLICY CONTEXT

2.1 The relevant Swindon Borough Local Plan 2026 policies are summarised below.

- Policy SD1: Sustainable Development Principles;
- Policy DE1: High Quality Design;
- EN1: Green Infrastructure Network; Policy
- EN2: Community Forest;
- EN3: Open Space;
- EN4: Biodiversity and Geodiversity Policy
- EN5: Landscape Character and Historic Landscape Policy
- NC3: New Eastern Villages – including Rowborough and South Marston Village Expansion;

2.2 In addition the New Eastern Villages (NEV) Green Infrastructure Supplementary Planning Document (SPD) is relevant.

3.0 PURPOSE, AIMS AND OBJECTIVES OF THE FRAMEWORK

3.1 The primary aim of this Framework is to ensure that the proposed development and future management of the Green Infrastructure preserves and enhances the landscape, biodiversity and arboricultural resources in the long-term.

3.2 This document draws upon the principles set out within the Environmental Statement (ES) and subsequent addendum, the Outline Biodiversity Enhancement Options plan[1] , which accompanied the outline application.

3.3 The Ecology Mitigation Strategy has therefore sought to integrate the requirements of the consented development, as identified in the Planning Application for the site with the existing ecological interest and legal obligations associated with protected species. The aims of this Strategy are outlined below.

3.4 The aims of this document are to:

- Identify green infrastructure for retention and opportunities for creation of new habitats;
- Identify potential ecological impacts on and relevant protection for the South Marston Brook in the context of the wider site strategy;
- Provide a strategy for new and retained wet features;
- Provide a strategy for delivering a biodiversity net gain;
- Identify opportunities to provide access to nature.

Biodiversity Net Gain (BNG)

3.5 The requirement for a BNG, a requirement for the FLEAMP Condition 10, has led the identification of habitats that will be provided as part of the site Green Infrastructure, providing an objectively led mechanism maximising gains within the site.

3.6 The UK Government, as signatory to the Rio Convention on Biological Diversity, is committed to conserving and enhancing biodiversity. This commitment is further enforced in the Natural Environment and Rural Communities Act (NERC) 2006 and the Natural Environment White Paper (June 2011).

3.7 DEFRAs 25 Year Environment Plan (2018) seeks to embed a ‘net environmental gain’ principle for development to deliver environmental improvements locally and nationally. Current policy is that the planning system should provide BNG where possible. Following the Environment Bill receiving Royal Ascent in November 2021 mandatory biodiversity net gain will be required under the Environment Act.

3.8 Part 6 of the Environment Act 2021 is entitled 'Nature and Biodiversity'. Within this part is section 98, entitled 'Biodiversity gain as condition of planning permission'. It states that 'Schedule 14 makes provision for biodiversity gain to be a condition of planning permission in England'. Part 1 of Schedule 14 contains the requirement for biodiversity net gain including the use of a biodiversity metric and the need for 10% BNG.

3.9 Section 147(3)(s) highlights that Part 6 of the Environment Act 2021 only comes into force 'on such day as the Secretary of State may by regulations appoint'. Thus, it is not until further secondary legislation is made by the Secretary of State does Part 6 of the Environment Act (where the 10% requirement exists) come into play.

3.10 In the absence of secondary legislation coming forward making the requirements of the Environment Act mandatory, the term “measurable net gain” is not defined within the NPPF.

[1] FPCR Environment and Design Ltd (2017) Outline Biodiversity Enhancement Options Drg. no.: 1796-L-03 rev D

Additionally, this term currently has no agreed definition in local or UK policy. Whilst a figure of 10% is widely viewed as best practice, it currently has no adopted policy support at either a local or national level, and is therefore considered an aspirational target, and not a mandatory requirement. It is noted that the outline consent does not include a condition that requires provision of a specific target in relation to biodiversity net gain.

3.11 The NPPF (2021) seeks to ensure that the planning system contributes to and enhances the natural and local environment, protect and enhance biodiversity and geodiversity by:

"174. d) minimising impacts on and providing net gains for biodiversity, including by establishing coherent ecological networks that are more resilient to current and future pressures;

179. b) promote the conservation, restoration and enhancement of priority habitats, ecological networks and the protection and recovery of priority species; and identify and pursue opportunities for securing measurable net gains for biodiversity."

3.12 This report accompanies an ES Chapter for the Site which has been undertaken to inform the development proposals and to provide recommendations for mitigation and enhancement (of which measurable biodiversity net gain will form a part of).

3.13 Natural England's published biodiversity net gain metric is an MS Excel spreadsheet that is used to quantify the predicted net-change in biodiversity value ("biodiversity units") of a proposed development site before and after development. It treats the flat "habitats" and linear features "hedgerows" and "rivers" separately, and is based on pre-determined values, along with published written guidance, set by a Natural England-led team of experts. The latest version of this metric is 3.1, published in April 2022, and this has been used for this scheme.

3.14 To facilitate this, the Site has been mapped and digitised using the Biodiversity Metric 3.1 QGIS Template, with the existing habitats identified and areas generated. In accordance with the 3.0 Metric User Guide, habitats have been defined under UK Habitat Classification by using the 'UKHab/Phase 1 translation' feature embedded within the Defra 3.1 calculator spreadsheets. The broad landscaping proposals for the Site were then uploaded into the QGIS template, and the proposed habitats mapped and digitised to generate areas for each of the habitats proposed for creation.

3.15 These pre- and post-development habitat areas were then inputted into the 3.1 Metric Calculation tool. Pre-development habitats were grouped into their habitat type and condition based on the results of the Extended Phase 1 Habitat and condition assessment surveys, while post-developments were classified into their UKHab type as identified from the proposed habitats within landscaping plans and given appropriate target condition scores. The metric then provides a habitat distinctiveness score for each of the baseline and proposed habitats which are pre-assigned scores based on the habitat type.

3.16 The strategic significance of the habitats was also assessed for both the pre- and post-development habitats based on the location of the site, its proximity to existing areas of biodiversity interest and its setting within wider habitat corridors.

3.17 The metric then assigns a range of pre-assigned factors to each of the proposed habitats. These have been advised by subject knowledge experts and are universal multipliers generated by the metric itself for the following variables relevant to habitat creation, enhancement, or restoration proposals:

- Difficulty of creating or restoring/enhancing a habitat: This pre-assigned score is based on how difficult a particular habitat type is to create or restore/enhance
- Temporal risk: This is the 'time to target condition' for any particular habitat and determines how long a particular habitat type is likely to take to reach the condition score that the desired condition score assigned to it.
- Spatial Risk: This score is based on the distance between the site of habitat loss and any habitats creation or enhancement proposals at any offsite offsetting solutions.

3.18 As a part of this process a Habitat Condition Assessment Survey was undertaken by Peter Hoy (FISC 5 botanical surveyor) and James Hutchison (FISC 4 botanical surveyor) and was completed in accordance with Natural England's The Biodiversity Metric 3.1 technical supplement. The guidance provides a list of criteria for a range of broad habitat types which are used to determine whether habitat fall into a 'poor', 'moderate' or 'good' condition score.

3.19 The corresponding Biodiversity Impact Assessment is provided at Appendix A.

3.20 Full details of the calculation methodology used is provided in the Biodiversity Metric 3.1 – User Guide .

4.0 GREEN INFRASTRUCTURE FRAMEWORK MANAGEMENT PROPOSALS

4.1 The site forms a parcel of agricultural land lying between the small village of South Marston to the north and the A420 and the Great Western railway line to the south. The site extends from Thornhill Road in the West to just beyond Old Vicarage Lane which runs along the eastern boundary.

4.2 The site is currently, largely in intensive agricultural use and comprises arable fields, including both cereal crops and temporary grass (turf) with some improved pasture. The field pattern is typical of the wider landscape with fields being generally irregular in shape and predominantly of small scale.

4.3 Hedgerows and ditches define the field pattern. A number of the hedges are tall and thick containing a number of mature trees, others are variable in quality, largely due to current management practices. As well as some seasonally wet ditches associated with hedge lines, there are some small field ponds within the site. Small areas of broad-leaved woodland occur in association with field ponds.

4.4 Trees were present throughout the site, were representative of a rural landscape where farming predominates and consisted of a diverse range of species. The existing trees are best described as being of several distinctive broad types, dependent upon the growing environment, position and function. The highest concentrations of more significant tree cover were the frequently occurring specimens contained within the field boundaries and the well treed belt aligning South Marston Brook. Incidental individual and groupings of trees were found associated with the farms, existing residential boundaries which abutted the site, field pond and other minor water / ditch courses, railway embankments and roadside locations. The condition of trees varied from good to poor, the latter being mainly due to an absence of any targeted management and age-related decline.

4.5 Where possible, opportunities will also be sought to ensure that the resident species are able to utilise retained and restored areas through the provision of a variety of habitats known to be of importance within the site. Habitat creation measures will endeavour to provide of a wide range of

environmental conditions and habitats that are known to be of value to specific groups and to more generally.

4.6 While specific measures will be detailed within each respective CEMP and phase LEAMP, broad species mitigation and enhancement aims will include:

- avoid unnecessary disturbance during site works through careful programming, implementation, and supervision,
- where opportunities arise aim to create habitats using general principles known to benefit specialist groups of interest

4.7 A summary of ecological receptors and constraints is provided in Table 1 below:

Table 1: Valued Ecological Receptors

Valued Ecological Receptor	Nature Conservation Significance	Rationale
Habitats		
Ponds	Local/Low	Some potential value and appreciably enriches the habitat diversity of the site
Mature trees	Local/Low	Some potential value and appreciably enriches the habitat and structural diversity of the site
Veteran trees	District	Habitat of principal importance under S41 NERC Act and uncommon resource in the wider area
Hedgerows	Local/Low	Semi-natural/undisturbed habitat that is likely to provide a habitat for a range of wildlife and potentially important dispersal function through the Site
Fauna		
Great crested newts	District	GCN are protected under the Wildlife and Countryside Act 1981 and Habitats Regulations 2010 and are included on the NERC Act S41 list. Small and declining populations. Relatively widespread species generally although localised.
Bat assemblage	District	Supports a typical suite of species characteristic of farmland habitats. Relatively rare species (barbastelle) recorded but uncommon and at low frequency All bats are protected under the Wildlife and Countryside Act 1981 and Habitats Regulations 2010 and are included on the NERC Act S41 list.
Reptiles	Local	Small population of grass snake which is partially protected and included on the NERC Act S41 list
Breeding and overwintering birds	Local	Supports a typical assemblage of breeding and over-wintering birds including the specially protected species barn owl and red kite, although breeding within the site unlikely.
Badgers	Local	Badger social group present in the wider area, which is protected under the Badger Act

Habitat Retention

4.8 Sensitive site design will prioritise retention of habitats providing functional connectivity across the site.

4.9 Existing ponds, woodland blocks, individual and groups of trees and hedgerows will be retained and protected where feasible. While these habitats are of nature conservation value, very little active management appears to have been undertaken for the benefit of nature conservation in recent years.

4.10 Enhancement work will bolster the existing ecological interest and provide additional habitat for species of interest known to be present within retained and disturbed habitats through appropriate management or intervention.

Habitat Creation

4.11 New habitat creation will aim to increase site-wide diversity, compensate for the loss of previous habitats; and provide connectivity to the wider landscape. Green infrastructure areas outside of urban areas (i.e. those with significant ecological value) are illustrated in figure 2.

4.12 Green infrastructure design will include a range of habitats, with native broadleaved tree planting, scrub and species-rich meadow grassland, to enhance the site's biodiversity and led by the Biodiversity Impact Assessment (BIA), with semi-natural habitats in close association to create a more varied mosaic in these areas, including adjacent to off-site habitats. Linear features including hedgerows, trees, grassland and scrub will provide corridors and stepping-stone habitats through the built areas. Attenuation features will be designed to maximise their biodiversity benefit, both in form and structure to provide organic non-uniform features that will support a range of habitats associated with wetlands, including aquatic and emergent vegetation, wet and marshy grassland and over-deepening to provide permanent water.

4.13 Newly created ponds will be designed to provide benefits to amphibians and other wildlife. The ponds will be shaped to provide a range of bank angles and heights. This will create differing conditions of light and temperature and will thus maximise and encourage the diversification in the flora and associated fauna. Margins will be shallow to maximise productivity and the potential for wildlife, and have shelves of low gradients, gently shelving to the pond base. Deeper water is highly desirable as this provides a buffering effect from cold air cooling or freezing of the upper layers of water than in shallow ponds, especially in spring; an important period for amphibian reproduction.

4.14 Grassland habitat creation will utilise species rich meadow seed mixes and will provide suitable habitat for wildlife including amphibians, reptiles, birds and invertebrates. The grassland habitat will form a mosaic with the scrub planting and will buffer the woodland areas. Grassland areas will be managed to promote structural diversity including tussocky grassland, mown pathways and clearings, and areas of bare ground, creating a range of micro-climates.

4.15 New structurally diverse scrub/woodland planting will increase habitat diversity and connectivity for woodland species through the site. The aim of woodland creation will be to promote development of a stratified structure with canopy, understory and ground flora. This will be achieved through specifically targeted Woodland creation will utilise a diverse range of mostly native broadleaved species alongside associated and appropriate management promoting diversity in both age and spatial structure. Retention of both standing and fallen deadwood will be promoted within woodland settings to contribute to the wider biodiversity. Depending on any specific management objectives for woodland areas, temporary clearings and coppicing if appropriate will allow greater light penetration and promote natural regrowth, increasing opportunities for butterflies and other insects, which in turn provide food for birds, bats and mammals and thus will be considered to meet policy aspirations.

4.16 Scrub habitat will be designed to form a mosaic with the grassland habitat. A diverse structure will include numerous clearings, glades and rides providing extensive edge habitats. Planting mixes will include a range of native species providing nectar, seeds, fruits, shelter and nest sites for invertebrates, birds and mammals.

4.17 Native hedgerow creation will compensate for any necessary losses in existing provision. New locations will be selected to promote cross-site connectivity. New native hedgerow planting on site is to be made in association with new ditches/banks. (This should be a minimum 0.5m ditch depth, with arisings used to form mound to elevate planting). There should be a stand off from the ditch to the hedge base of at least 1.5m-2m. Ditches should have inclined banks avoiding vertical steps.

4.18 Gaps in hedgerows created to facilitate the access road will be reinforced by providing “hop-overs”. These will be designed to minimise any potential impact to foraging/commuting routes connecting to the wider landscape. The planting will include the implementation of standard trees adjacent to the road which will grow to be above the level of vehicle movement.

4.19 Where habitats are exposed to increased disturbance or restricted management due to proximity to urban areas, planting mixes will favour a diverse range of native species. Ensuring there is a varied range of seasonal resources will provide maximum benefit for wildlife.

4.20 Additional artificial habitat creation such as nest and roost boxes, hibernacula, log piles and basking banks, will supplement green infrastructure provision.

4.21 Standard tree planting will be carried out in areas of public open space, as part of landscape buffers and residential areas not only to create visual amenity but to provide increased local canopy cover and habitat. These trees will provide habitat for local bird populations and other species dependant on trees.

Habitat Management

Wetland Habitats

4.22 Extant wetlands and ponds within the site are, arguably, the most important features within the site due to the varied plant, amphibian, reptile, bird and invertebrate life that they support. Management works may require a Natural England licence, either through the District Level Licensing (DLL) route or traditional licensing process, if there is a risk of disturbance to great created newts.

4.23 Some enhancement work to existing ponds will be required through over-deepening and the thinning/removal of over shading vegetation, although, following this, existing ponds are likely to be relatively self-sustaining, with little management likely to be required in the early years. Longer-term management of these areas will be dynamic to each situation. Regular, gentle thinning-out of excess aquatic vegetation will prevent the build-up of nutrients.

4.24 Specific areas of scrub control may be required early on in the management of the site where it is currently over-shading areas of water edge habitat. Scrub from these areas should be removed using brush-cutters where practical. Similarly, overshading outgrown willow pollards, will benefit from re-pollarding.

4.25 Pond creation will be designed and located to provide stepping-stone habitat for amphibians. New ponds will be surrounded by good quality terrestrial habitat including species-rich grassland, woodland blocks and a network of scrub. The waterbodies will be shaped to provide a range of bank angles and heights. Gradients will vary from 15º - 25º from horizontal and will be enhanced

by the excavation of small embayments. This will create differing conditions of light and temperature and will thus encourage diversification in the flora and associated fauna. Variations in water depth will be created from shallows to deeper pools to enhance nature conservation value and to maximise the development of biodiversity. The shallowest areas will grade into an expanse of seasonally wet mud that may attract a variety of invertebrates and plants, which will in turn attract other fauna including birds and mammals

4.26 In order to maintain a range of pond types, the successional process can be reset by re-profiling areas or digging new basins. Such management should be targeted at areas of low ecological interest.

4.27 New hibernation and refuge features, including brash / log piles and artificial hibernacula, will be created in association with new and existing ponds. These features will be sited to maximise the availability of basking and refuge habitat they provide.

4.28 Egg laying heaps will be provided for grass snake. Heaps will be constructed by piling vegetation such as grass clippings, leaf litter, or woodchips on top of a base of brash. The brash will create spaces within the heap to allow easy access to nesting females. It will also increase aeration, aiding decomposition of the organic material and boosting the heat generated.

Future management principles

4.29 Retained ponds will be managed to maintain their existing condition. Newly created ponds will be manged to achieve moderate condition. The following management criteria for woodland ponds should be targeted:

- the pond is not artificially stocked with fish,
- the pond water levels should be able to fluctuate naturally throughout the year,
- the pond is not artificially connected to other waterbodies, either via streams, ditches or artificial pipework,
- there is an absence of non-native plant and animal species,
- less than 10% of the pond is covered with duckweed or filamentous algae.

4.30 For non-woodland ponds, management will target all of the above criteria and also the following additional criteria:

- the surface of non-woodland ponds is no more than 50% shaded by woody bankside species.

4.31 Where clearance is required, this should be undertaken by hand and all arisings will be left on the bank of the pond for a 2-week creep back period, allowing any invertebrates or amphibians time to return to the water prior to the removal of the arisings from site.

Trees

4.32 The mature trees are features of high biodiversity value in the context of the site and the immediate surrounds. Significant amounts of arboricultural / tree work would be highly likely to reduce their value for biodiversity. The removal of areas of rot, decay, cavities and deadwood in the interests of safety of local residents using the green spaces would remove habitat niches that are important for assemblages of associated species which depend on large mature trees including those species which are legally protected such as nesting birds and roosting bats.

4.33 Any tree surgery will be considered as a 'last resort' where there are no alternative 'safe' options to be adopted. Any works that would be required will be subject to the required pre-works bat surveys.

4.34 It is recognised that trees are important for several reasons and therefore management objectives should avoid any unnecessary arboricultural management and tree works and maintain a careful balance between keeping the public safe, retaining amenity and minimal loss of any biodiversity value.

4.35 Existing trees will require management in the future. Where existing mature trees are retained within Green Infrastructure, these trees will only be subject to arboricultural management where it is either imperative for public safety or there are clear benefits for wider biodiversity. Footpaths within green space with public access will be designed to be set at appropriate distances so that they can be maintained with retention of deadwood, cavities and decay as these features will be important for the whole tree ecosystem.

4.36 Older trees will naturally develop deadwood and cavities which increase their biodiversity value over time. Assessment of tree health, disease and features of biodiversity value will form part of any targeted management plans with the trees located in spaces that will be accessible for inspections and any tree surgery if required.

4.37 In addition to existing trees, new trees will be incorporated into proposals in areas of proposed parkland, such as the South Marston Country Park. These will provide additional structural diversity to open areas and provide steeping stone habitats through development parcels, along both Primary Streets (e.g. Parkland Way) and Secondary Streets

4.38 The management principles will be set out as follows:

<u>Existing Retained Individual Trees – future management principle</u>

- Any immediately necessary remedial work to existing trees and hedgerows on site will be undertaken prior to the occupation of dwellings. A review as to whether there are any specific requirements for management of existing trees on the site will be undertaken.
- Within publicly accessible areas, and especially within proximity of defined footpath routes within green space, trees would be checked annually, after major storms and at any other interval deemed necessary following inspection for essential remedial works with works undertaken as advised by a suitably qualified Arboriculturist.
- All works will be undertaken according to best practice as set out in BS 3998 *Tree Work – Recommendations* and using qualified tree surgeons/or arborists to satisfy a duty of care.
- All vegetation and, particularly, woody vegetation proposed for clearance should be removed outside of the main bird-breeding season (March - July inclusive) as all birds are protected under the Wildlife and Countryside Act, 1981 (as amended) whilst on the nest. Where this is not possible, vegetation should be checked for the presence of nesting birds by a suitably qualified Ecologist.
- Arisings from tree works will, where practical, be used to create small dead wood piles within the woodland planting. Alternatively, where this is not possible, they can be chipped to provide mulch for use in amenity planting areas or removed from site.

- Standing dead wood will be retained in situ where possible to provide habitat. Where it is not safe to retain this, the dead wood piles and recumbent logs will be created and placed at marginal areas, providing habitat for a range of fauna including saproxylic invertebrates and fungi, plants, small mammals, common reptiles and amphibians.

New Tree planting

4.39 Detailed descriptions of management operations including watering, topping up mulch levels, fertilising, pruning, stake and guard maintenance are set out below:

- For at least the first year, trees will be watered at least weekly during periods of low or no rainfall. They should be watered to field capacity each time via irrigation tubing. During the first 5 years following planting trees will be watered in periods of extreme drought (2 or more weeks without substantial rainfall). After establishment continue to water only if deemed to be required.
- Top up mulch levels where necessary.
- Fertilise new areas of tree planting using an approved liquid feed (N10:P15:K10) at a rate of 60g/ m^2 during early May and late September.
- Prune back any diseased or rotten wood (including the removal of main stems and limbs) back to sound wood.
- Examine tree stakes and ties for their effectiveness and requirement. If the tree has yet to establish, replace or adjust ties, spacers and tree tubes as appropriate. Well the tree has established well, then all stakes, ties, spacers, tubes etc will be removed and disposed of at a licensed landfill site and holes, surfaces made good.
- Any trees that fail within the first 5 years will be replaced by the management company at their own cost. (Trees failing within the first year will be replaced by the contractor who has installed them)
- Any tree that dies or is necessarily felled as a result of storm damage, vandalism, mechanical damage, fungal decay or other such cause, but which is not felled or removed as part of any programme of thinning or coppicing, shall be replaced like for like by the management company or by the appointed contractor installing them if within the first year. Such replacement would be with a tree of either the same species or similar species as those existing to allow some flexibility and to avoid problems encountered with 'Same Tree Disease' to ensure a sustainable tree cover in the interests of visual amenity.

Woodland

4.40 In addition to the above, newly planted woodland will be provided to buffer existing woodland habitats and enhance connectivity by creating stepping-stone woodland blocks and belts. Scrub planting will be used to create a graded woodland edge habitat adding further value to the woodland belts.

4.41 Retained and newly planted woodland will be managed to achieve moderate condition by targeting the following criteria:

- trees have a diverse age structure,

- trees are protected from herbivore damage,
- woodland to be maintained free of invasive species, including rhododendron and laurel,
- woodland composition to include at least 5 native species,
- tree cover in upper (>5m) and understorey (up to 5m) layers dominated by native species,
- woodland includes areas of natural regeneration and where appropriate coppicing,
- trees are healthy and free of disease/pests,
- woodland has a diverse vertical structure, with multiple distinct or a complex of storeys,
- standing and fallen deadwood >20cm diameter and >1m tall/long present within the woodland,
- woodland is free of significant nutrient enrichment/damage.

4.42 While some early management may be required to ensure successful establishment, including the measures outlined for trees above, and some thinning in the medium term, it is intended that retained areas will be allowed to largely develop naturally, maintaining a good variety of tree and shrub age classes and the humid conditions required by some of the lower plants.

Scrub

4.43 Scrub will be located throughout the GI. Areas of more restricted extent in proximity to developments will be targeted to reach moderate condition, whilst larger areas will target good condition.

4.44 To achieve moderate condition the following criteria will be targeted by management:

- there is a good age range – all of the following are present: seedlings, young shrubs and mature shrubs,
- there is an absence of invasive non-native species (as listed on Schedule 9 of WCA, 1981) and undesirable species make up less than 5% of ground cover,
- habitat is representative of UKHab description (where in its natural range). There are at least three woody species, with no one species comprising more than 75% of the cover (except common juniper or box, which can be up to 100% cover).

4.45 To achieve good condition management will target all of the above criteria and also the following additional criteria:

- there are clearings, glades or rides present within the scrub, providing sheltered edges,
- the scrub has a well-developed edge with scattered scrub and tall grassland and/or herbs present between the scrub and adjacent habitat(s).

4.46 Scrub is of greatest value to wildlife where it forms a scattered patch work/network, which provides ample edge habitat. Therefore, scrub planting should be maintained to ensure that it does not encroach into more open habitats. Native species planting will provide a food source, shelter and nesting opportunities for invertebrates and birds. Management should seek to maintain a variable age structure, through rotational cutting and thinning.

4.47 Management operations associated with clearings, rides and glades should seek to ensure continued availability of a wide range of conditions and niches suitable to maintain the lifecycles of the species present, providing a diverse interface between habitats. The ride centre should be cut

every year or two to maintain grassy or herbaceous vegetation, whereas the edges could be cut on a longer rotation (e.g. 3-7 years) to encourage a dense thicket to develop although the exact timing should be dependent on site conditions and development. Cutting bramble and shrub vegetation at the edges of rides at varying angles to the vertical would also increase the overall surface area.

Grassland Habitat

4.48 A mixture of grassland habitat types and conditions are envisaged across the site. Each broad grassland type will require differing habitat creation and management prescriptions.

Modified Grassland

4.49 Modified (low distinctiveness) grassland will be located in areas urban areas such as those utilised for sports and play which will be subject to high levels of human disturbance and more regular management. To achieve moderate condition, management will target the following criteria:

- some scattered scrub may be present, but scrub accounts for less than 20% of total grassland area,
- cover of bracken less than 20%,
- there is an absence of invasive non-native species (as listed on Schedule 9 of WCA, 1981) and undesirable species make up less than 5% of ground cover,
- physical damage evident in less than 5% of total grassland area, such as excessive poaching, damage from machinery use or storage, damaging levels of access, or any other damaging management activities.

4.50 Management of this amenity grassland type through regular cutting would seek to ensure a sward height of between 70mm and 150mm during the growing season to ensure resilience to drought and avoid the need for chemical fertilizers and herbicides.

Other Neutral Grassland

4.51 Other neutral grassland will be located throughout the wider GI. Restricted areas in proximity to developments will be targeted to reach moderate condition, whilst more open areas will target good condition.

4.52 To achieve moderate condition the following criteria will be targeted by management:

- there must be 9 or more species per m2,
- cover of bracken less than 20% and cover of scrub (including bramble) less than 5%,
- there is an absence of invasive non-native species (as listed on Schedule 9 of WCA, 1981). Combined cover of undesirable species and physical damage (such as excessive poaching, damage from machinery use or storage, damaging levels of access, or any other damaging management activities) accounts for less than 5% of total area.

4.53 To achieve good condition management will target all of the above criteria and also the following additional criteria:

- cover of bare ground between 1% and 5%,
- sward height is varied (at least 20% of the sward is less than 7 cm and at least 20 per cent is more than 7 cm), although this would be dependent on when monitoring occurs,
- the appearance and composition of the vegetation closely matches characteristics of the specific UKHab grassland habitat. Wildflowers, sedges and indicator species for the specific grassland habitat type are very clearly and easily visible throughout the sward.

4.54 In order to achieve moderate condition and provide sufficient species diversity and a greater diversity of flowing herbs initial soil testing will be undertaken to identify soil nutrient status, which will help inform the future management /monitoring strategy. Where soil phosphate levels are high and monitoring indicates that excessive growth of more vigorous /coarse species is preventing the establishment of other species a more intensive cutting regime will be implemented in the early years of establishment to reduce soil nutrient status, followed by the localised use of yellow rattle to aid in reducing the vigour of more competitive grasses.

4.55 Fields will be cultivated by either the removal of vegetation/ploughing and harrowing then seeded with an appropriate wildflower seed mix or green hay if available. The mix would reflect soil conditions and will be detailed within the LEAMP.

4.56 If green hay is to be used it should be cut just before the hay stage and collected and spread immediately, prior to wilting (ideally within an hour or two). Taking hay at this time means it contains a higher numbers of seeds from a range of species, instead of being grass dominant.

4.57 Hay should be spread thinly and evenly, with the sward visible beneath the hay. Once spread, the hay should be left for at least one week in dry weather, or three weeks in wet weather, to allow the seed to fall. After this period sheep should be allowed to graze the site in order to trample the seeds into the soil.

4.58 A sensitive and appropriate long-term management strategy is essential for the success of sward enhancement. In the initial year this is likely to include more regular cutting between July and November, to keep the sward short and aid germination. Cutting should again be implemented in the first spring to avoid seedlings being shaded out by developing vegetation. Perennial weeds should be controlled early on by spot treatment of herbicide where necessary.

4.59 In subsequent years, the treated grassland compartments should be managed as traditional hay meadows to maintain species diversity. Management can be expected to include cutting in late July and subsequent swath turning to assist with seed-shedding. Perennial weeds should be continued to be treated with herbicide. No inorganic fertilisers are used. Herbicide use will be limited to spot treatment.

4.60 Where soil tests are found to be particularly high in phosphate levels, the first year/s may require more intensive management with monitoring of the grassland more regular to fine tune management.

4.61 Bare ground is an important component of good condition grassland habitats. It provides suitable areas for the colonisation of stress tolerant but uncompetitive species and an important resource for invertebrates and reptiles, especially when located close to areas of cover. Grassland management should include the cyclical creation of new areas of bare ground as existing areas become colonised by vegetation.

4.62 Management in the short to medium term will incorporate a varied cutting regime in addition to periodic disturbance in key locations. This should also seek to ensure there is adequate provision of early successional states and bare ground within the whole mosaic of habitats. Where cutting is proposed the standard method would be to allow species to complete their life cycle before cutting i.e. flowering and setting seed. Varying the timing of the cut can have a significant effect on species composition. As a general rule earlier cutting removes more rank vegetation at the most productive time of the year. Later cutting enhances diversity in nutrient poor areas.

4.63 In the first years of establishment, created grassland would be monitored to determine the need for and extent of any management required to reduce the cover of invasive ruderal species. Following establishment of the grassland it would be necessary to cut the area at least once a year with the removal of all arisings. Some of the arisings should be used to create refuges and egg laying substrate for reptiles at strategic locations.

4.64 Provision for the potential for grazing of some areas could be made in the form of suitable fencing and through the provision of water, where necessary, allowing some degree of flexibility in future long-term management and to maximise the benefits of grassland management for biodiversity.

SUDS

4.65 The primary function of the surface water drainage system will be to control flows and reduce the potential for pollution of the aquatic environment. However, it is important that SUDS are considered holistically and that each element is recognised as part of a network of habitats and wildlife corridors.

4.66 Prior to discharging to the South Marston Brook, surface water will discharge to a mixture of swales and ponds/attenuation basins situated on the peripheries of built development parcels. Drainage lagoons are usually installed at an early stage of construction works. This will, for the later phases of development, have allowed their natural colonisation by plants and animals and therefore offer water quality improvements for the discharge during remaining construction works.

4.67 Principles followed will include:

- Where possible, SUDS will be located close to but not immediately directly connected to existing wetland areas, so plants and animals can naturally colonise the new SUDS ponds.
- Well vegetated shallow bays will be created allowing the establishment of areas of marsh and open water.
- Smoothly finished surfaces would be avoided; although they give the impression of tidiness they provide less micro habitat diversity for plants and animals.
- Ponds and attenuation basins should be designed to hold permanent water throughout the breeding season (March-July) and ideally throughout the year in most years.
- If planting is essential, such as for the earlier phases ensure only native plants of local origin are used.

4.68 Sustainable urban drainage features will be designed and managed to enhance biodiversity value. To achieve a condition score of moderate, the following criteria will be targeted:

- invasive non-native species (Schedule 9 of WCA) cover less than 5% of total vegetated area,
- the water table is at or near the surface throughout the year.

4.69 Management of attenuation basins should aim to enhance the value for nature conservation complementing their function for flood risk management and include the following:

- Tall emergent vegetation should not dominate the pond to the point where it shades out other less dominant plants, as this will reduce the diversity of the pond.
- Around the ponds there should be areas of rough tussocky grassland, meadow or tall grass field margin and hibernacula/refugia/egg laying heaps for amphibians/reptiles.
- The appointed contractor will be responsible for the detection of and control/removal of invasive species at the watercourses e.g. New Zealand Pygmyweed *Crassula helmsii*, Parrot's Feather *Myriophyllum aquaticum* and Indian Balsam *Impatiens glandulifera* etc, if become present. Current Environment Agency and Natural England guidance will be consulted to obtain details of the most up-to-date methods of treatment of an invasive species.
- The appointed contractor will be responsible for the detection of great-crested newt (*Triturus cristatus*) prior to works commencing as this species has been recorded in ponds within 500m of the site. Should at any time great-crested newt become be suspected or found within new attenuation basis, no management or maintenance works should be undertaken until further advice is provided by a suitably experienced ecologist.

4.70 Long term management of attenuation basins should consider the following to minimise impacts to nature conservation:

- Emergent species will not be permitted to encroach over the attenuation basins to maintain at least 75% open water where basins hold permanent water.
- Removal of vegetation in permanently wetted regions suitable amphibian habitat will only be carried out outside the amphibian breeding season (typically March – June and outside the amphibian hibernation period (typically November-February) if habitat is suitable to support hibernating amphibians and outside the nesting bird season (March – August) All removed material will be left in small piles around the basin margins for two to three days prior to removal from site, this will permit aquatic invertebrates and any over-wintering larvae to return to the basin.
- Clear shingle perimeters and drainage layers, removing unwanted vegetation outside the amphibian and bird breeding period and outside amphibian hibernation period i.e. undertaken September/October each year as appropriate/required.
- Check inlets/outlets/sediment traps and remove any blockages. ▪ Remove litter and other inorganic debris as required.

Hedgerows

4.71 The interface between the development edges and the wider Green Infrastructure has an important role in defining the character of the development, delivering ecological benefits as well as contributing to a safe and attractive public space.

4.72 In order to provide some definition between public and private space, native hedgerows would be planted to provide an interface between and into built development parcels with appropriate gaps to allow pedestrian access through to the wider green infrastructure. Timber post and rail/estate fencing would be placed on the development side of the hedge.

4.73 Hedgerow creation will compensate for any loses required to facilitate development. The following condition criteria should be targeted to achieve good condition:

- Height >1.5m average along length,
- Width >1.5m average along length,
- Gap between ground and base of canopy <0.5 m for >90% of length,
- Gaps make up <10% of total length and no canopy gaps >5 m,
- >1 m width of undisturbed ground with perennial herbaceous vegetation for >90% of length measured from outer edge of hedgerow, and is present on at least one side of the hedge,
- Plant species indicative of nutrient enrichment of soils dominate <20% cover of the area of undisturbed ground,
- >90% of the hedgerow and undisturbed ground is free of invasive non-native and neophyte species,
- >90% of the hedgerow or undisturbed ground is free of damage caused by human activities.

4.74 To achieve a thick hedge, planting will comprise five native plants per metre, in double staggered rows.

4.75 The hedgerows will be cut back once per annum for the first 5 years between December and January. Once the hedgerows have developed a bushy habit, they will be maintained on rotation with no more than 50% of any hedgerow being cut in any one year. Hedgerows will be maintained at a height between 3-4m, as and when required hedgerows will be laid or coppiced in order to maintain dense cover at the base providing important habitat for wildlife.

Orchard Planting

4.76 The potential for Orchard planting is included within the Design Code. Maiden form apple trees *Malus domestica* using local cultivars/or those agreed with SMBC will be planted between October and March, avoiding periods of inundation or prolonged frost. This will accord with BS 8545:2014. Trees are to be mulched using wood chippings or bark to establish a 1m diameter around the tree stem.

4.77 Formative pruning will be undertaken whilst the trees establish (initial 5-10 years as required) in accordance with advice provided in the Natural England Technical Information Note TIN016[2] to begin developing open frameworks of strong lateral branches. Following the formative stage the main focus of pruning will change to one of general maintenance and encouraging fruit production, in accordance with Technical Information Note TIN017[3].

4.78 Routine monitoring and care as per 'Tree Planting' above.

4.79 Grassland habitats below orchard trees would be treated as per Neutral grassland as above.

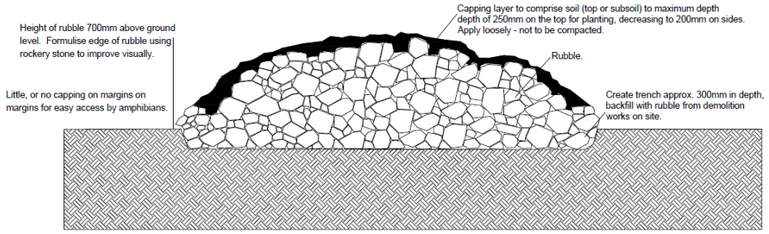
Detailed Framework Management Prescriptions

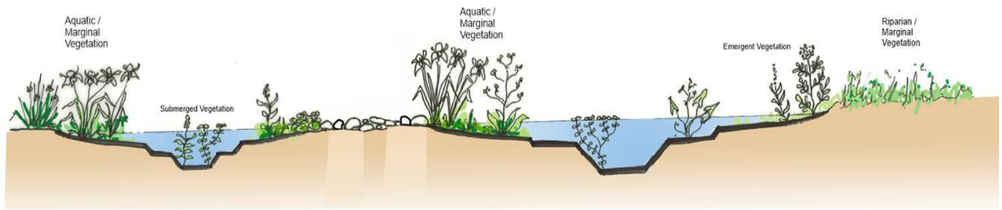
[2] Natural England 2010. Technical Information Note TIN016. Traditional orchards: formative pruning of young trees
[3] Natural England 2010. Technical Information Note TIN017. Traditional orchards: maintenance pruning.

4.80 Detailed framework management prescriptions on which subsequent LEAMPS/Reserved Matters applications will be based are provided in Table 2 below

Table 2: Detailed Framework Management Principles

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION

1a	**Wetland**
	Create and manage new water bodies/attenuation basins for use by breeding amphibians and other wildlife while maintaining function as attenuation basins.
		Pond design:
		• Located to provide stepping-stone habitat for amphibians – within 200m of nearest neighbour new/existing ponds.
		• New ponds will be surrounded by good quality terrestrial habitat including species-rich grassland, woodland blocks and scrub. Each pond will include two hibernacula.
		Indicative Hibernacula Specification:
		
		Plan to show Hibernacula - NOT TO SCALE
		 
		
		• The waterbodies will be shaped to provide a range of bank angles and heights.
		  ○ Gradients will vary from 15º - 25º from horizontal and will be enhanced by the excavation of small embayments. This will create differing conditions of light and temperature and will thus encourage diversification in the flora and associated fauna.
		  ○ Variations in water depth will be created from shallows to deeper permanent pools to enhance nature conservation value and to maximise the development of biodiversity.
		  ○ The shallowest areas will grade into an expanse of seasonally wet mud that may attract a variety of invertebrates and plants, which will in turn attract other fauna including birds and mammals.
			Establish and maintain conditions for wildlife
			Year 1 after over-sowing
			• Post sowing (first growing season) – control flush of annual weeds by mowing regularly and removing the clippings.
			• Maintain a sward height of 40-75mm.
			• Do not apply any herbicide or fertiliser.
			
			Thereafter - annual management.
			• Leave un-cut or periodic cut and removal of emergent vegetation to provide no more than 65% vegetation cover. Management to promote broad margin of emergent vegetation around permanent water/drawdown zone.
			• Management/planting to promote emergent vegetation around inflow headwall (to help filter pollution/suspended material) including supplementary planting if deemed necessary
			• Annually inspect for scrub development and hand-pull any scrub saplings to maintain at least 50% unshaded.
			• Monitor habitat quality and take remedial steps to prevent or treat pollution/eutrophication such as use of barriers/filters in event of upstream pollution event or removal of severe algal bloom/duckweed.
			• Periodic trimming of bankside vegetation at least once every three years as necessary.
			• Not to be stocked with fish. Periodic monitoring and removal of non-native/fish introductions/other non-native animals± and Schedule 9 plants where evident

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
		Indicative Pond Cross Section:  New planting should include native species and undertaken as specified within the LEAMP for each reserved matter application. This could be expected to include the preparation of a well consolidated seed-bed which could be sown with Emorsgate EP1F Wildflowers for Wetlands seed mix (or similar) at a rate of 1.5g/m2, ideally mid-September to end of 1st week in October. Achillea ptarmica — Sneezewort Lycopus europaeus — Gypsywort Angelica sylvestris — Wild Angelica Lythrum salicaria — Purple Loosestrife Caltha palustris — Marsh Marigold Mentha aquatica — Water Mint Eupatorium cannabinum — Hemp Agrimony Pulicaria dysenterica — Common Fleabane Filipendula ulmaria — Meadowsweet Ranunculus acris — Meadow Buttercup Geum rivale — Water Avens Scrophularia auriculata — Water Figwort Hypericum tetrapterum — Square-stalked St John's-wort Iris pseudacorus — Yellow Iris Lotus pedunculatus — Greater Bird's-foot-trefoil Scutellaria galericulata — Skullcap Silene flos-cuculi — Ragged Robin Succisa pratensis — Devil's-bit Scabious Vicia cracca — Tufted Vetch	
1b	**Wetland** Restore existing retained ponds to provide a habitat for use by wildlife including reptile and amphibians	• Reinstate willow pollarding and remove scrub from within pond • Remove excessive silt build-up/over-deepen to allow retention of permanent open water	As above, although shaded margins can extend to over 50% due to wooded nature of existing pond.
2	**Neutral (species-rich) grassland** Create and maintain a species rich grassland of benefit to willdlife	• Remove existing vegetation and prepare fine, consolidated seedbed. • Spread locally sourced green hay or sow suitable seed mix. The mix would reflect soil conditions and will be detailed within the LEAMP e.g.: Emorsgate EM5 Meadow Mixture (or similar) at a rate of 4g/m^2 mid-September to end of 1st week in October. **Wild Flowers** *Achillea millefolium* — Yarrow; *Centaurea nigra* — Common Knapweed *Daucus carota* — Wild Carrot; *Galium verum* — Lady's Bedstraw *Geranium pratense* — Meadow Crane's-bill; *Knautia arvensis* — Field Scabious	Implement traditional management Year 2 after sowing • From June to mid-September leave un-mown to flower and set seed freely. • From mid-September to end of May mow regularly and remove clippings. ○ Arisings to be relocated, in part, to designated areas on the southern boundary • Do not apply any fertiliser.

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		Lathyrus pratensis Meadow Vetchling *Leucanthemum vulgare* Oxeye Daisy *Malva moschata* Musk Mallow *Plantago media* Hoary Plantain *Prunella vulgaris* Selfheal *Rhinanthus minor* Yellow Rattle *Trifolium pratense* Red Clover *Leontodon hispidus* Rough Hawkbit *Lotus corniculatus* Common Bird's-foot-trefoil *Plantago lanceolata* Ribwort Plantain *Primula veris* Cowslip *Ranunculus acris* Meadow Buttercup *Rumex acetosa* Common Sorrel **Grasses** *Agrostis capillaris* Common Bent *Anthoxanthum odoratum* Sweet Vernal-grass *Festuca ovina* Sheep's Fescue *Festuca rubra* Slender-creeping Red-fescue *Briza media* Quaking-grass *Cynosurus cristatus* Crested Dog's-tail *Phleum bertolonii* Smaller Cat's-tail *Trisetum flavescens* Yellow Oat-grass • Post sowing (first growing season) – control flush of annual weeds by mowing regularly and removing the clippings. • Mow when sward height reaches 100mm; do not mow lower than 50mm.	• Spot treat any undesirable ruderal species such as docks and thistles using a contact herbicide applied via a hand-held weed wipe or applied with a hand brush – do not spot treat with a knapsack sprayer. AND/OR in combination with: Year 2 onwards – annual management • Graze with cattle from end of August until sward no longer provides sufficient food for cattle without supplementary feeding, or ground conditions become too wet to graze without poaching. • Ensure all livestock removed by end of March, then allow free growth until end of August. • No herbicide or fertiliser inputs.
3	**Standard Trees (inc orchard trees)** Maintain and extend standard tree cover across the site	New tree planting to be in line with the Strategic Design Code: Tree Strategy, detailed as specified within the for each reserved matter application • Any immediately necessary remedial work to existing trees and hedgerows on site will be undertaken prior to the occupation of dwellings. A review as to whether there are any specific requirements for management of existing trees on the site will be undertaken. • Arisings from tree works will, where practical, be used to create small dead wood piles within the woodland planting. Alternatively, where this is not possible, they can be chipped to provide mulch for use in amenity planting areas or removed from site.	• Within publicly accessible areas, and especially within proximity of defined footpath routes within green space, trees will be checked annually, after major storms and at any other interval deemed necessary following inspection for essential remedial works with works undertaken as advised by a suitably qualified Arboriculturist. • All works will be undertaken according to best practice as set out in BS 3998 Tree Work – Recommendations and using qualified tree surgeons/or arborists to satisfy a duty of care. • All vegetation and, particularly, woody vegetation proposed for clearance should be removed outside of the main bird-breeding season (March - July inclusive) as all birds are protected under the Wildlife and Countryside Act, 1981 (as amended) whilst on the nest. Where this is not possible, vegetation should be checked for the presence of nesting birds by a suitably qualified Ecologist. • Arisings from tree works will, where practical, be used to create small dead wood piles within the woodland planting. Alternatively, where this is not possible, they can be chipped to provide mulch for use in amenity planting areas or removed from site. • Standing dead wood will be retained in situ where possible to provide habitat. Where it is not safe to retain this (e.g when outside of Primary/secondary Streets or, in some cases, Development Parcels), the deadwood piles and recumbent logs will be created/ placed at marginal areas, providing habitat for a range of fauna including saproxylic invertebrates and fungi, plants, small mammals, common reptiles and amphibians.
4	**Woodland and scrub** New woodland and scrub will be allowed to grow to maturity and, in time, scrub to succeed to woodland where desirable	New tree and shrub planting should include native species and undertaken as specified within the LEAMP for each reserved matter application native broadleaved and conifer tree planting and native scrub species. Species for inclusion within strategic green space to be chosen carefully to ensure they are appropriate for the area and will contribute to the overall aims of strategy to provide a net gain in biodiversity value. Broadleaved woodland planting mixes should favour the inclusion of oak, birch, hazel and field maple as appropriate for the National Character Area. Typical species lists are outlined below: Broadleaved Woodland *Acer campestre* Field Maple (15%) Alnus glutinosa Alder (5%) *Betula pendula* Silver Birch (10%) *Betula pubescens* Downy Birch (10%)	Early promotion of canopy species including weed management and understorey scrub management. • Tree shelters and stakes to be inspected twice annually as part of the general maintenance visits and adjusted accordingly. Damaged ties or stakes must be replaced. When the trees are established and can support themselves the ties should be carefully removed and the stakes removed/cut down to ground level. This operation is likely to be required after three to five years dependant on establishment rates, stability and growing conditions. • Periodic monitoring and control/removal of non-native species as necessary, including any Schedule 9 species. • Replacements of like-for-like in event of failure or promotion of neighbouring established specimens as deemed necessary.

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
		Corylus avellana Hazel (15%) Fagus sylvatica Beech (5%) Pinus sylvestris Scots Pine (5%) *Quercus petraea* Sessile Oak (15%) *Quercus robur* English Oak (15%) Taxus baccata Yew (5%) <u>Wet Woodland</u> Alnus glutinosa Alder (30%) *Betula pendula* Silver Birch (10%) *Betula pubescens* Downy Birch (25%) Salix caprea Goat Willow (10%) Salix cinerea Grey Willow (25%) <u>Scrub</u> *Acer campestre* Field Maple (10%) *Cornus sanguinea* Dogwood (10%) *Corylus avellana* Hazel (12.5%) *Crataegus monogyna* Hawthorn (20%) Cytisus scoparius Common Broom (2.5%) *Quercus robur* English Oak (2.5%) *Rhamnus cathartica* Alder Buckthorn (12.5%) *Rosa canina agg.* Dog-rose (10%) *Sambucus nigra* Elder (10%) *Viburnum opulus* Guelder Rose (10%) A suitable woodland mix, such as the Emorsgate EW1 Woodland mix or similar agreed, will be surface sown in autumn or early spring onto a prepared seed bed. To support the establishment and long-term success of new planting the following measures should be adopted: • Trees and shrubs are to be checked annually when they are in full leaf to ensure that they are thriving. Defects requiring remedial works will be recorded. • Any trees and shrubs which die or do not establish will be replaced the first planting season following each annual inspection during the first five years after planting • Cut areas of scrub in a rotation, aiming to retain all ages. Scrub typically matures in 15 years, so cut 1/5 every third year. • Newly planted trees and shrubs to be strimmed in a sensitive manner, to avoid damage to any habitats to be retained, from around the base of the plants until the point it is determined that they are tall enough to tolerate weed competition. If the trees and shrubs show signs of poor growth or reduced vigour an application of the appropriate fertiliser should be carried out. If trees and shrubs do not respond to a treatment of fertiliser, further inspections are required, including the ground conditions for signs of compaction, contamination, and poor-quality topsoil. Any problems uncovered with the soil must be remediated. Should remediation works not resolve the problem a replacement tree may need to be planted. • During establishment trees should be watered regularly during prolonged dry periods. • Newly established woodland ground flora will produce more growth and grasses will be more prominent. In these conditions an annual cut mid summer may be worthwhile for a more managed appearance and to keep weeds of semi-shade such as nettles and brambles in check. After around 10-15 years or once the tree canopy has closed little management would be required.	• All tree guards, tubes, ties and stakes that have not biodegraded naturally are to be removed from site once plants are mature enough to survive browsing. Periodic trimming/management of understorey scrub in planting groups including: • Regular cutting to create dense stands with graded margin (at woodland/glade edge); and • Infrequent management of selective stands to promote more open stands (including thinning of stands and light trimming to aid succession of canopy species). • Plantings protected with rabbit guards as necessary. • Monitoring and installation of fencing or other protection as deemed necessary. Periodic monitoring and control of browsing pressure. • In every management year: • Retention of trimmings following management (where not diseased) in brash piles and windrows. • Retention of standing deadwood including branches and whole trees (where not diseased).

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
		• All vegetation removal should be timed outside the bird nesting season, i.e. undertaken between September and February. Should vegetation removal be required outside this timeframe an ecologist should be consulted prior to works commencing. • Shrubs are vulnerable to browsing by livestock, deer and rabbits during establishment. Protect with spiral guards or tubes (using non-plastic and biodegradable where possible) and are removed when the plants are mature enough to survive browsing.	
5a	**Hedgerows** To maintain & enhance existing native species hedgerows	• Protect existing hedgerows during construction in accordance with the CEMP and BS 5837:2012, or as superseded.	Implement sympathetic management • Cut 50% of each length on a rotational basis between January and February (so that autumn fruits are available for wildlife) and again avoiding frost periods. • Cut to no less 2m high. • Retain a minimum 1.5m margin at base and cut every 3 years to promote dense cover. • Laying hedge once new plantings have established between November and the end of February to maintain a dense structure
5b	**Hedgerows** – native species (new hedgerows within the development)	Create new hedgerows using species which reflect the species-composition of the existing hedges • Prepare vegetation-free planting strip. • Plant using staggered double row, 500mm between rows at a rate of 6 plants/m. • Plant 60-90cm bare root whips. • Planting mix to reflect species composition of existing hedges eg : <u>Native shrubs</u> *Acer campestre* Field Maple (10%) *Cornus sanguinea* Dogwood (5%) *Corylus avellana* Hazel (10%) *Crataegus monogyna* Hawthorn (50%) *Lonicera periclymenum* Honeysuckle (5%) *Rhamnus cathartica* Buckthorn (5%) *Rosa canina agg.* Dog-rose (5%) *Quercus robur* English Oak (10%) • Protect with shrub guards • Seed with a suitable hedgerow seed mix such as Emorsgate EH1 Hedgerow mix or similar approved	Implement sympathetic management • Upon establishment (year 2-3) cut 50% of each length on a rotational basis between January and February (so that autumn fruits are available for wildlife) and again avoiding frost periods. • Cut to no less 2m high. • Retain a minimum 1.5m margin at base and cut every 3 years to promote dense cover. • Laying hedge once new plantings have established between November and the end of February to maintain a dense structure

5.0 SPECIES-SPECIFIC MEASURES

5.1 A number of protected species and/or otherwise notable species are known to occur locally. As a result, some unavoidable disturbance/displacement of some species will occur. Nevertheless, in order to ensure that disturbance is kept to a minimum, appropriate measures based on best practice techniques would be employed to prevent unnecessary or adverse impacts arising. Specific measures during construction will be secured by adherence to a CEMP.

5.2 Where possible, opportunities will also be sought to ensure that the most important species are able to utilise retained and enhanced areas, either through direct translocation or through the provision of a wide variety of habitats known to be of importance within the site. Specific management prescriptions would be informed by ongoing and future protected species surveys and detailed within each LEAMP for that Phase. Nevertheless, habitat creation measures will endeavour to provide of a wide range of environmental conditions and habitats that are known to be of value to specific groups.

5.3 Species-specific measures are outlined below, although would be supplemented by ongoing/future surveys of each phase/Reserved Matters Application site boundary.

Bats

5.4 The potential for impacts to bats from the use of lighting within any construction/built development areas would be mitigated by the sensitive design which will follow published best practice guidance. Measures to reduce impacts will include:

- The use of directional floodlighting to avoid spill;
- Avoidance of potential dispersal corridors;
- The use of the lowest intensity possible;
- All luminaires should
  a. lack UV elements when manufactured. Metal halide, fluorescent sources would not be used.
  b. LED luminaires should be used where possible due to their sharp cut-off, lower intensity, good colour rendition and dimming capability.
  c. A warm white spectrum (ideally <2700Kelvin) should be adopted to reduce blue light component.

Existing Trees

5.5 Where felling operations are required either during the construction phase or as a result of routine monitoring/arboricultural surveys within the Green Infrastructure that identify remedial measures the following will be followed:

- Inspection by the licensed bat worker
- The removal of ivy, where present, and re-inspection for cavities/holes/splits
- The section felling of limbs/trunks where cavities/holes/splits occur
- Final inspection by the bat worker

Enhancement

5.6 Bat boxes will be erected as part of the enhancement strategy.

5.7 Exact locations to be confirmed at Reserved Matters through the associated LEAMP (condition 11) but to include a range of boxes and locations. Bat bricks will be installed in new dwellings (at a rate of 1 in 5 dwellings for each separate development parcel, and in accordance with BS 42021:2022 British Standard for integral nest boxes, or as superseded) in suitable locations as directed by the ecologist. Where ecologically suitable locations are constrained by the development location, any excess bat brick provision will be carried over to be utilised in subsequent development parcels.

5.8 Across the whole South Marston development area, an additional 80 bat boxes (woodcrete or other durable material), including those allocated as part of specific mitigation strategies, will be installed on suitable retained trees (e.g. 2F Schwegler Bat Box (General Purpose) or similar. 1FW Bat Hibernation Box or similar). The final location of the boxes will be provided on an annotated plan to Swindon Borough Council for approval and provided to the appointed maintenance contractor.

5.9 The inspection of the bat boxes may cause the disturbance of bats, a UK and European protected species, and will therefore only be undertaken by an appropriately licensed and experienced surveyor. The bat boxes will be monitored and checked annually where they fall within the control of the management contractor in September/October (i.e. outside the bird breeding season). Subject to any relevant ecological constraints, damaged or missing boxes will be replaced within three months of the defect being identified

Birds

5.10 To avoid disturbance to breeding birds, where vegetation is to be removed it will be removed prior to the bird-breeding season (March to August/Sept). If this is not possible, vegetation will be checked prior to removal by an experienced ecologist. If active nests are found, vegetation will be left untouched until all birds have fledged. Specific advice will be provided prior to undertaking the clearance. This would be a statutory requirement due to the protection of all nesting birds and their nests under the Wildlife and Countryside Act, 1981. A suitably qualified ecologist would supervise this.

5.11 If an active bird nest is found in vegetation or other feature/habitat to be removed, the area would be retained and clearance operations would recommence only when the nest is no longer in use.

5.12 A range of bird nesting opportunities would also be created within the site. Proposals will include:

- In-built swift bricks (the provision of which will equal the total number of dwellings within each separate development parcel), in minimum clusters of three as sited by an ecologist and in accordance with BS 42021:2022 British Standard for integral nest boxes, or as superseded.
- Additional provision of nest boxes within the South Marston development wide green infrastructure on retained trees and other features including:
    a. To encourage nesting by small hole nesting species, a total of sixty boxes including a mixture of small nest boxes with holes from 25mm to 28mm in diameter. These should be sited 1-5m above the ground on trees with a clear flight path to the entrance.
    b. Twenty small open fronted (aperture of 100mm) boxes are likely to be used by robins and will encourage nesting by spotted flycatchers and other species. These should be sited on walls or trees 2-4m above the ground. Spotted flycatchers are most likely to take to these

boxes if they are situated within a climber such as ivy with a clear open view. These types of boxes may occasionally be used by other species such as blackbirds and wrens.

c. Two specialist nest boxes for stock doves comprising a large aperture (approx. 130mm). These can be attached to trees but should be sited at least 3m above the ground, have a clear flight path to the entrance and be in an area that is normally free of human activity.

d. One specialist nest box for kestrels comprise a large open fronted box usually rectangular in shape. These can be attached to trees but should be sited at least 5m above the ground.

e. Two specialist boxes for barn owl are readily available and are best placed on mature tree standards away from major roads. Boxes placed below the crown where they are most visible are most likely to be seen by passing owls and will not be obscured around the entrance. A height in the region of 4.5-7m may be achieved depending upon the tree concerned. If no suitable trees are present, barn owls boxes maybe erected on poles, however the design of pole mounted boxes are different to those mounted on trees.

5.13 In general, all bird boxes should be sited avoiding prevailing wind and rain, south-east is generally a preferred direction. Locations of all nest box types will be decided upon by an experienced ecologist who can advise the most appropriate location depending upon the species involved and their requirements in terms of surrounding habitat and favoured nest sites. This will ensure that the boxes receive the best chance of occupancy.

Badger

5.14 Badgers use of the site appears to be confined to foraging. It will therefore be necessary to ensure that access through the site is still available.

5.15 In accordance with best practice, a full badger survey will be conducted prior to the clearance of any area. This will identify the presence of any newly excavated setts that may be lost and / or disturbed during site clearance. Where any recently excavated setts are identified and disturbance impacts are considered significant, mitigation will be put in place for any affected setts and may be ratified under licence by Natural England. This is likely to involve either supervised and licensed sett closure and/or the use of protective badger fencing where appropriate.

5.16 Additional measures considered good practice would also include suitable construction site management such as ensuring that retained habitat is appropriately fenced to prevent inadvertent disturbance and badgers are prevented from entrapment where deep excavations are likely to occur within their range or close to important routes of movement.

Reptiles

5.17 The availability of suitable habitats is variable, with large areas of short vegetation and agricultural habitats being unsuitable. These habitats are not considered likely to support over wintering reptiles due to a lack of suitable ground conditions and absence of potential refugia and could be cleared during winter.

5.18 Areas of greater suitability occur scattered throughout the site where more structural diversity in the vegetation and close juxtaposition of areas of shelter and basking occurs such as along hedgerows and locally in the north eastern extent of the site where less disturbed habitats occur.

The removal of reptiles from working areas will utilise a range of methods based on the suitability of the habitat to be affected and season:

a) Clearance of vegetation over winter, where no suitable over-wintering/hibernacula habitat occurs

b) Passive displacement, where clearance of small, isolated areas of homogenous vegetation or suboptimal habitat is proposed

c) Trapping and translocation will be undertaken where more extensive habitats occur and the likelihood of reptiles being present greatly reduced.

5.19 At the time of writing, suitable habitat that will be lost and may require trapping and translocation is extremely limited and as such it is not considered necessary to create a specific receptor site. It is acknowledged that, this may change in the interim prior to the development of some parcels. Given this uncertainty a specific receptor site has not been identified. Nevertheless, where it is considered necessary a receptor will be created in the south of the south adjacent to the southern boundary/rail line, where suitable habitat will be created.

5.20 Enhancement to the site would include the replacement of extensive agricultural land with limited suitability for grass snake, with a structurally varied mixture of open grassland, scrub and woodland habitats providing the close juxtaposition of areas of shelter, foraging and basking necessary for this group and detailed within the respective LEAMP. Additional features including hibernacula/egg-laying sites created using the arisings from grassland management and ponds would also provide additional habitat and habitat for prey items.

Hazel Dormouse

5.21 Dormice are arboreal species spending most of their life above ground within hedges, scrub, trees and woodland. During winter months, dormice hibernate on the ground in shallow depressions, leaf litter, log piles, root bases of hedgerows and within grassy margins along suitable vegetation.

5.22 Dormice surveys in 2022 did not detect any evidence of occupation within the site boundary. A single historical record from 2013 was noted within 2km of the site. While suitable habitat is present on site, limited local records, and the negative survey results indicate the likely absence of dormice. As such, the hazel dormouse is not considered a constraint to the development.

5.23 The following measures will be adopted to promote the retention and creation of suitable dormouse habitat, ensuring hazel dormouse are able to commute through the site and enable the potential expansion of dormouse ranges onto the site.

- Existing retained hedgerow should be safeguarded with appropriate protective measures to BS 5837:2012 as set out in the Arboricultural Method Statement.
- New native hedgerow planting within the green infrastructure will occur across the site at a density and width to provide sufficient cover for dormouse in accordance with the prevailing good practice guidelines for this species[4]. This density of planting is considered to provide sufficient cover and protection to minimise the risk of predation by domestic cats.
- New native hedgerow planting should contain species of suitable foraging value for hazel dormouse such as Oak, Honeysuckle., Dog Rose and Hawthorn. Retained hedgerows should

[4] Bright, Morris & Mitchell-Jones (2006) The Dormouse Conservation Handbook (2nd edition).

be supplementary planted with these species where possible to enhance the retained hedgerows for dormouse using the species listed above.

Great Crested Newts

5.24 Compensation for any effects on great crested newts would be achieved through off-site enhancements funded by District Level Licensing compensatory payments.

5.25 Enhancements to the site would also include the replacement of extensive agricultural land, some with limited suitability for great crested newts, with a structurally varied mixture of open grassland, scrub woodland and new wetland habitats within the broader Green Infrastructure providing a range of shelter and foraging opportunities for this group.

5.26 Additional features including hibernacula and refuge sites created using the arisings from grassland and woodland tree management. Would be provided close to existing and new ponds providing shelter for GCN. It is anticipated that two features would be provided in association with each pond with their specific locations would be detailed within the respective LEAMP.

Riparian Mammals – Otter and Water Vole

5.27 No suitable habitat will be affected during the South Marston phase. Measures for the enhancement and management of habitats associated with the South Marston Brook will be detailed within the Rowborough FLEAMP.

6.0 BIODIVERSITY NET GAIN

6.1 Under the NPPF development must achieve a biodiversity net gain and provide habitats which are of greater diversity and interest than those to be lost, ensuring habitat corridors are maintained or created around boundaries and through the site into the wider area. The proposals will maintain and enhance existing ecological networks and links in the local area, including with the habitat corridor following the South Marston Brook to enhance connectivity with the wider ecological network. A full assessment of South Marston Brook will be included as part of the Rowborough FLEAMP.

6.2 Biodiversity Net Gain has then been used to inform the habitat creation and enhancement proposals for the scheme and to guide decisions around additional habitat provision. The condition assessment criteria provide a set of measurable objectives, against which management can be assessed, reviewed and, where necessary, modified as part of a review process.

6.3 All new and retained habitats will be subject to sensitive on-going management to promote their biodiversity and viability in the long term. A detailed 30-year biodiversity management plan (LEAMP), based on the detailed framework management principles set out above, will be produced to support this as part of the Reserved Matters Application necessary for each phase, including details of the roles, responsibilities and management mechanism to be put in place. This document will be reviewed and updated at intervals of no greater than 5 years by a suitably qualified and experienced ecologist, arboriculturist and landscape architect, working collaboratively. Modifications to the management plan should ensure management practices remain optimum to achieve and maintain stated goals.

6.4 Ecological management of the site will continue beyond the initial 30 year period, throughout the operational lifetime of the development. The Plan and the management regime will be informed by the ongoing works at the site with reviews every five years in perpetuity to ensure the objectives and prescriptions are in accordance with current wildlife legislation and conservation objectives.

6.5 In order to ensure that habitat created within the site reaches and maintains its maximum value to nature conservation, habitats should be monitored regularly by a qualified ecologist.

Phasing

6.6 The South Marston site has been assessed as part of the wider Eastern Villages development to provide an overall net gain in biodiversity units for both area-based habitats and hedgerows.

6.7 To ensure that appropriate and sufficient Biodiversity Net Gain is delivered as part of each phase, the BNG assessment has been completed for each phase, both alone and in sequence, such that at any one point sufficient BNG can be delivered to avoid any deficit upon the completion of development at that phase.

6.8 A summary of each phase BNG, calculated using the Biodiversity 3.1 metric is provided in Table 2 below. The cumulative total tracks South Marston and Rowborough areas separately, with a combined total provided for the site wide calculations. The Metric (Appendix A) and accompanying plans (figures 3-6g) can be used as a framework to identify minimum habitat areas to be provided. The Metric should however be regarded as an iterative document, being updated as each Phase/Reserved Matters comes forward, and in the event there are any changes to the phasing of the development.

Table 2: Cumulative Biodiversity Net Gain

		South Marston		Rowborough				Site Wide
		Phase 1	Phase 2	Phase 1	Phase 2	Phase 3	Phase 4	
Cumulative Total	Baseline (units)	126.73	179.79	149.42	183.41	239.86	284.55	464.34
	Post-intervention (units)	149.27	183.1	199.24	221.13	261.77	343.82	526.94
	% Change	**17.79%**	**1.84%**	**33.34%**	**20.56%**	**9.14%**	**20.83%**	**13.48%**

7.0 ACCESS TO NATURE

7.1 Public access to nature will be balanced against the need to protect sensitive receptors.

7.2 The Great Western Community Forest objectives of creating and maintaining a high-quality environment for local people by diversifying land-use, enhancing biodiversity and providing new opportunities for leisure, recreation, cultural activity, education, healthy living and social and economic development will be followed where possible.

7.3 A network of formal cross site pedestrian/cyclist routes is to be facilitated by several pathways which connect the existing local footpath network to key locations within the site. Pathways will provide access from residential areas across the GI and connecting to the surrounding area. The

pathways will provide a varied user experience, passing through the range of habitat types, providing a balance of openness, transition, and enclosure.

7.4 The provision of site wide furniture will facilitate ease of navigation, opportunities for relaxation and adequate hygiene whilst being sympathetic to the site's rural context. Information boards will highlight key biodiversity features providing educational opportunities for users. Sensitively placed play areas will encourage outdoor activity and guide pedestrian traffic to targeted locations.

7.5 Existing landscape features have been retained where possible within the development and will be protected through the operational phases. For any habitats assessed as vulnerable to disturbance, management will ensure that public access, lighting and noise will be kept to a minimum. Informal access into sensitive areas of green Infrastructure would be prevented by the construction of appropriate fencing and hedgerow provision.

7.6 To ensure that the landscape features within the site reach and maintain their maximum recreational, educational and aesthetic value, all landscape features are recommended to be monitored annually by a qualified landscape architect.

8.0 RESPONSIBILITIES, MONITORING & REVIEW

Responsibilities

8.1 Taylor Wimpey, Hallam Land Management and Hannick Homes will be responsible for future management and their appointed contractors will be responsible for the implementation of actions during construction period.

8.2 After construction, the Green Infrastructure will be handed over to a Management Company or other appropriate body. The LEAMP, supported by the detailed landscape scheme will be the mechanism through which roles and responsibilities will be defined.

Review of management plan

8.3 This Plan is a reviewable document, which can be updated and revised as required. The Plan and the management regime will be informed by the ongoing works at the site. The Plan will also be reviewed Years 1, 2 and 5, with reviews every five years in perpetuity to ensure the objectives and prescriptions are in accordance with current wildlife legislation and conservation objectives.

8.4 A monitoring and review programme will enable modifications to the Management Plan to be discussed, evaluated and updated. It is proposed that the review of the Management Plan will involve a steering group of the relevant stakeholders and consultees, including developer, Swindon Borough Council, Parish Council/resident representatives and other interested parties.

9.0 FRAMEWORK POST CONSTRUCTION AND MANAGEMENT PROGRAMME

Habitat/ Feature	Regular Operations/Works (Years 1-5)	Anticipated Frequency of Visits/Operations (see note below)												Frequency, notes
		J	F	M	A	M	J	J	A	S	O	N	D	
Wetland (created)	Creation & planting													
	Maintenance/replacement													To year 5
	Management – cutting and mechanical removal													Cutting on rotation 1/3 to ½ every 2-3 years – removing arisings. Mechanical removal as needed to maintain 1/3 to ½ open water
Wetland (Existing)	Re-pollard over hanging willow													Year 1 and every 5-6 years thereafter
	Management – cutting and mechanical removal of marginal vegetation/scrub													Cutting on rotation 1/3 to ½ every 2-3 years – removing arisings. Mechanical removal as needed to maintain 1/3 to ½ open water
Woodland	Planting													Year one only
	Seeding with woodland mix													Year one only
	Maintenance (including individual trees)													Annually to year 5
	Selective thinning & trimming a sinuous edge													Year 5 onward (thinning), 1/3 to ½ on rotation Every 3-5 years.
	Control of pernicious weeds/undesirable ground cover													as necessary subject to monitoring.
Scrub	Planting													Year one only
	Maintenance													Annually to year 5
	Selective thinning & trimming a sinuous edge													From year 2 – to promote regeneration of seedlings and saplings

Habitat/ Feature	Regular Operations/Works (Years 1-5)	Anticipated Frequency of Visits/Operations (see note below)												Frequency, notes
		J	F	M	A	M	J	J	A	S	O	N	D	
	Control of pernicious weeds/undesirable ground cover													Annually as needed to Year 5 then thereafter as determined by monitoring
Neutral Grassland/wet grassland (attenuation margin)	Cutting													In late summer/ autumn unless monitoring suggests additional cutting required due to excessive vigor
	Scrub/weed control													Year 1 as required for weed control (including in spring), otherwise annually from year 2
Hedgerow (new)	Planting													Year 1 only
	Seeding with hedgerow mix													Year 1 only
	Cutting													From year 3 and as necessary subject to monitoring
Hedgerow (existing)	Cutting													
Bat and Bat box	inspection and replacement where required													Years 1-5
Reptile/amphibian hibernacula	Inspection and remediation where required													Years 1-5
General	Site walkover to assess establishment success and any remedial actions													Years 1-5
	Litter pick													Monthly or as required
	Repair or remediation of hard landscaping													As required

Monitoring Feature	Specific monitoring objective	Year 1	Year 2	Year 3	Year 4	Year 5	Target
1. Mixed Scrub	Have proposed new scrub areas been planted? Has the recommended planting mix been used? Are the guards in place? Have dead plants been replaced if less than 75% establishment of original planting? Has management maintained rides, glades and a graduated edge structure?	✓ ✓ ✓	 ✓ ✓	 ✓ ✓	 ✓	 ✓	New scrub habitat with a diverse species and spatial structure. Forming a graduated edge habitat when in association with woodland areas.
2. Woodland	Have proposed new woodland areas been planted? Has the recommended planting mix been used? Are the guards in place? Have dead plants been replaced if less than 75% establishment of original planting? Has management maintained rides and glades?	✓ ✓ ✓	 ✓ ✓	 ✓ ✓	 ✓	 ✓	New woodland habitat with a diverse species and spatial structure.
3. Ponds	Have ponds been created? Have the recommended seed mixes been sown? Has the first-year management been implemented? Are the ponds holding water? Are the ponds free of litter? Are the ponds covered with less than 10% duckweed or filamentous algae?	✓ ✓ ✓ ✓	 ✓ ✓ ✓	 ✓ ✓ ✓	 ✓ ✓ ✓	 ✓ ✓ ✓	Ponds present and free of algal blooms and litter, and supporting a wetland flora.
4. Species-rich neutral grassland (created)	Has the wildflower meadow been sown? Has the recommended seed mix been used? Has the first-year management been implemented? Has long-term management been implemented?	✓ ✓ ✓	✓ ✓	✓ ✓	✓ ✓	✓ ✓	Presence of good quality, species-rich neutral grassland. Species diversity and structure to meet specific criteria as defined in relative LEAMP.

Monitoring Feature	Specific monitoring objective	Year 1	Year 2	Year 3	Year 4	Year 5	Target
5. Hedgerows – native species (existing)	Have gaps in retained hedges been planted? Has the recommended planting mix been used? Are the guards in place? Have plants been replaced if less than 75% establishment of original planting? Has biannual trimming taken place?	✓ ✓ ✓	 ✓ ✓ ✓	 ✓ ✓ ✓	 ✓ ✓	 ✓	Gap between hedges no longer present. Hedgerows remain intact, they are 2.5m high and have a flat-topped A-shape.
6. Hedgerows – native species (new hedgerows within the development)	Have proposed new hedges been planted? Has the recommended planting mix been used? Are the guards and mulch-mats in place? Have plants been replaced if less than 75% establishment of original planting? Has biannual trimming taken place?	✓ ✓ ✓	 ✓ ✓	 ✓ ✓	 ✓ ✓	 ✓	New hedgerows intact they are 2.5m high and have a flat-topped A-shape.
7. SUDS – maximise biodiversity value of attenuation basins	Have attenuation basins been created? Have the recommended seed mixes been sown? Has the first-year management been implemented? Are the basins free of scrub and litter?	✓ ✓ ✓ ✓	 ✓	 ✓	 ✓	 ✓	Attenuation basins present, free of scrub and litter, and supporting a wetland flora.
8. Annual monitoring	Has annual inspection taken place?	✓	✓	✓	✓	✓	Brief inspection report produced
9. Review of Management Plan	Has Management Plan been reviewed at end of 5th Year?					✓	Next 5 year Management Plan has been produced.